

Notes

- **Assignment**

- ◆ 2.10
- ◆ 2.11
- ◆ 2.22 (b) (e)
- ◆ 2.25
- ◆ 2.28 (a) (c) (e) (g)

- **Tutorial questions (Week 6)**

- ◆ Basic Problems with Answers 2.20
- ◆ Basic Problems 2.29
- ◆ Advanced Problems 2.40, 2.43, 2.47

That is ...

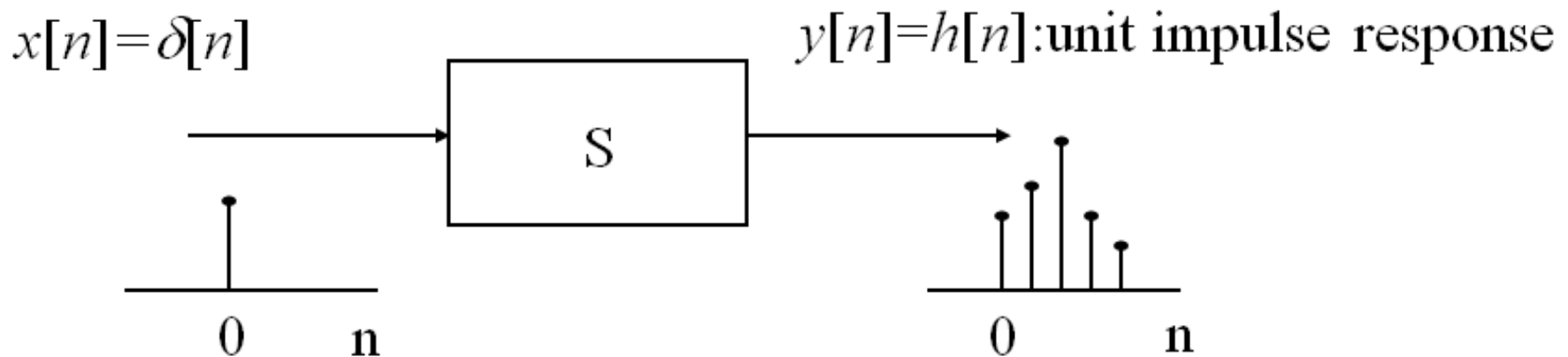
$$x[n] = \dots + x[-2]\delta[n+2] + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + \dots$$

⇓

$$x[n] = \sum_{k=-\infty}^{+\infty} \underbrace{x[k]}_{\text{Coefficients}} \underbrace{\delta[n - k]}_{\text{Basic Signals}}$$

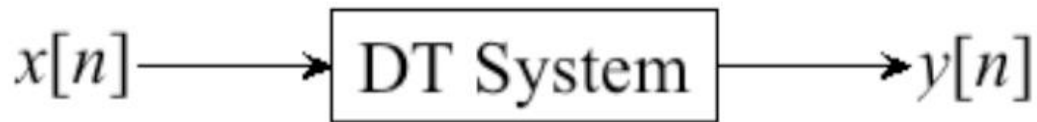
Important to note the “-” sign

Unit Impulse Response



Example: $y[n] = x[n] + 2x[n-1] + 4x[n-2]$
 What is unit impulse response?

Response of DT LTI Systems



- Now suppose the system is **LTI**, and define the *unit impulse response* $h[n]$:

$$\delta[n] \longrightarrow h[n]$$



From **T**ime-**I**nvariance:

$$\delta[n - k] \longrightarrow h[n - k]$$

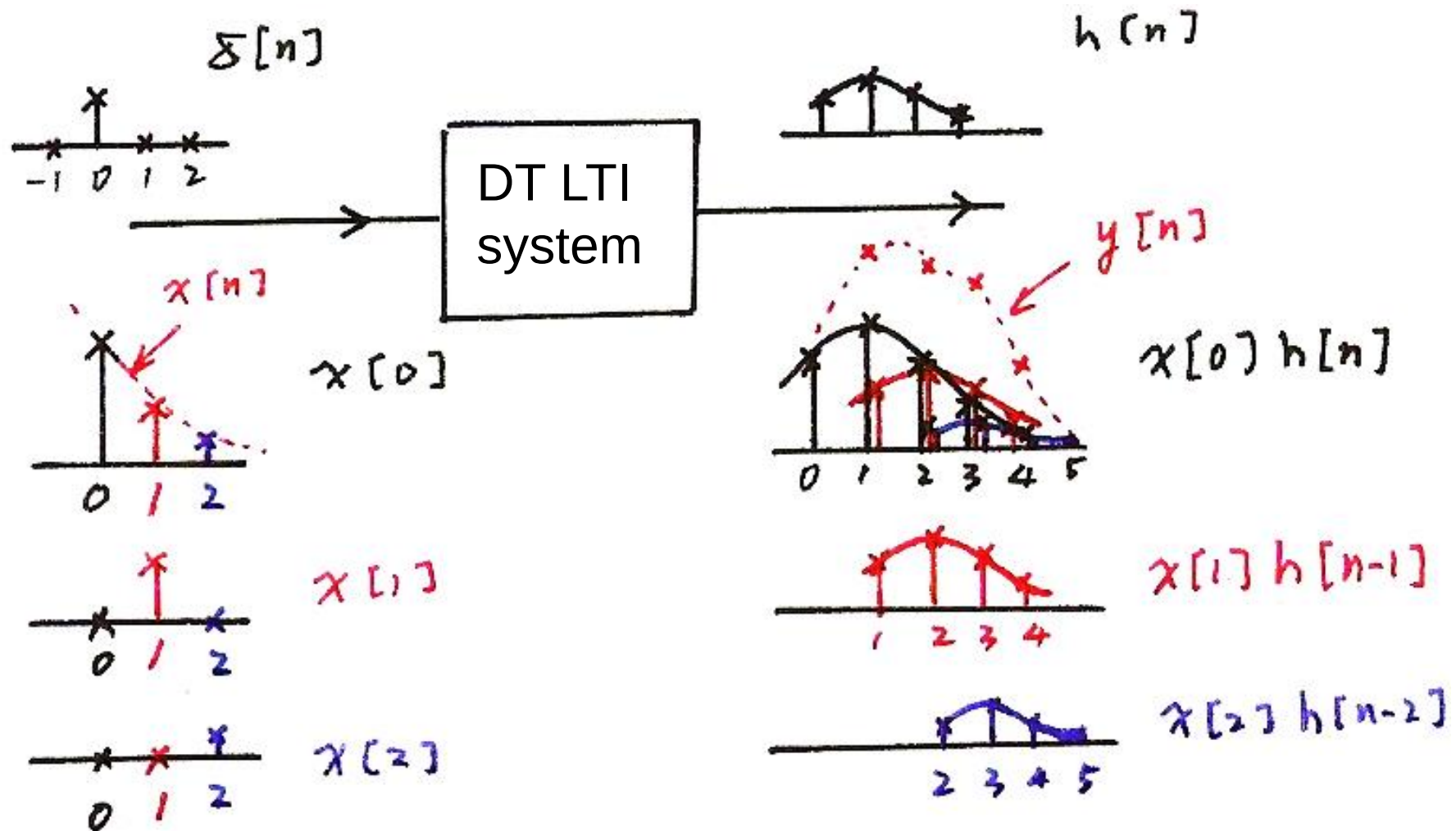
From **L**inearity:

$$x[n] = \sum_{k=-\infty}^{+\infty} x[k] \delta[n - k] \longrightarrow y[n] = \underbrace{\sum_{k=-\infty}^{+\infty} x[k] h[n - k]}_{\text{convolution sum}} = x[n] * h[n]$$

The output for an input signal is the superposition of a series of “shifted, scaled unit impulse response”

Chapter 2 Review

Input/Output Relation



Chapter 2 Review

- A different way to visualize the convolution sum
 - looked at on the index k

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Contribution to the output signal at time n

input signal

flipped version of $h[k]$ located at $k = n$

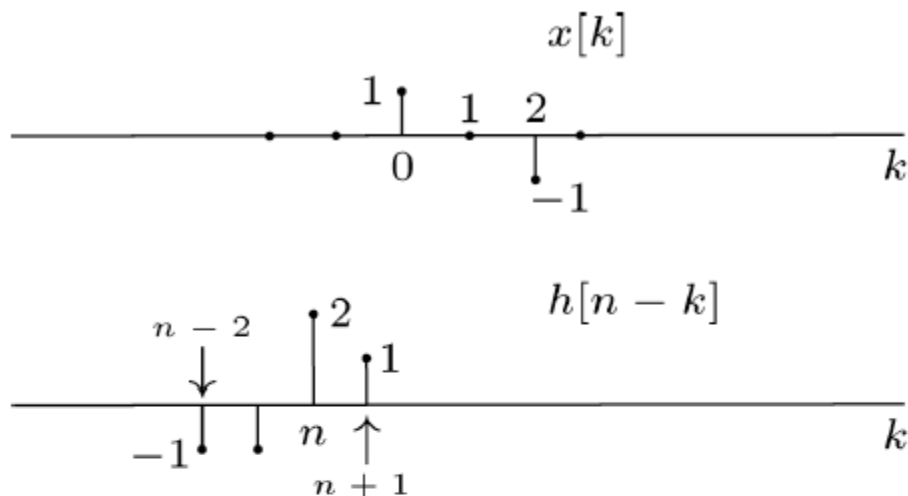
Convolution operation procedure:

$$\begin{array}{ccccccc}
 h[k] & \xrightarrow{\text{Flip}} & h[-k] & \xrightarrow{\text{Slide}} & h[n-k] & \xrightarrow{\text{Multiply}} & x[k]h[n-k] \\
 & & & & & & \xrightarrow{\text{Sum}} & \sum_{k=-\infty}^{\infty} x[k]h[n-k]
 \end{array}$$

F S M S

Calculating Successive Values: Shift, Multiply, Sum

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$$



$$y[n] = 0 \quad \text{for } n <$$

$$y[-1] =$$

$$y[0] =$$

$$y[1] =$$

$$y[2] =$$

$$y[3] =$$

$$y[4] =$$

$$y[n] = 0 \quad \text{for } n >$$

The Unit-impulse function $\delta(t)$

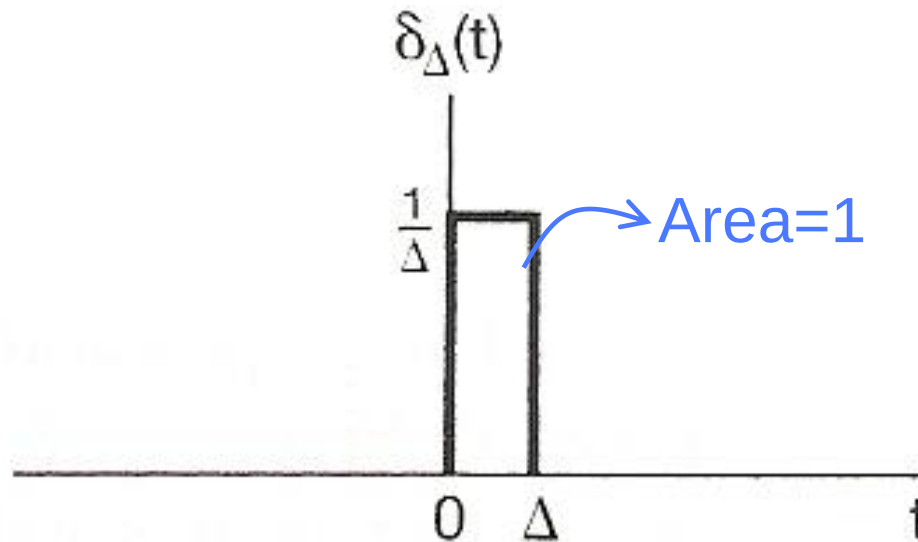
$$\left. \begin{array}{l} \delta(t) = 0 \quad \text{for } t \neq 0 \\ \delta(t) = \infty \quad \text{for } t = 0 \end{array} \right\} \quad (1)$$

$$\int_{0^-}^{0^+} \delta(t) dt = 1 \quad (2)$$

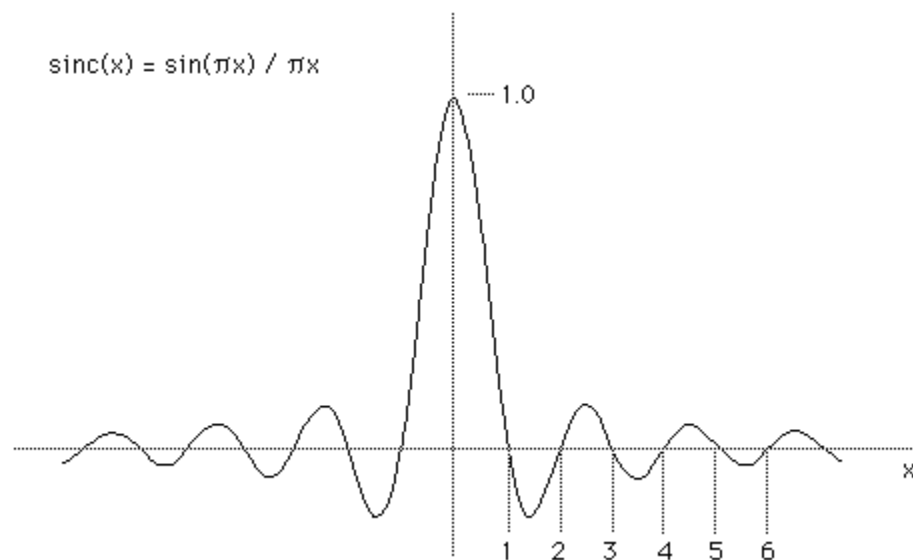
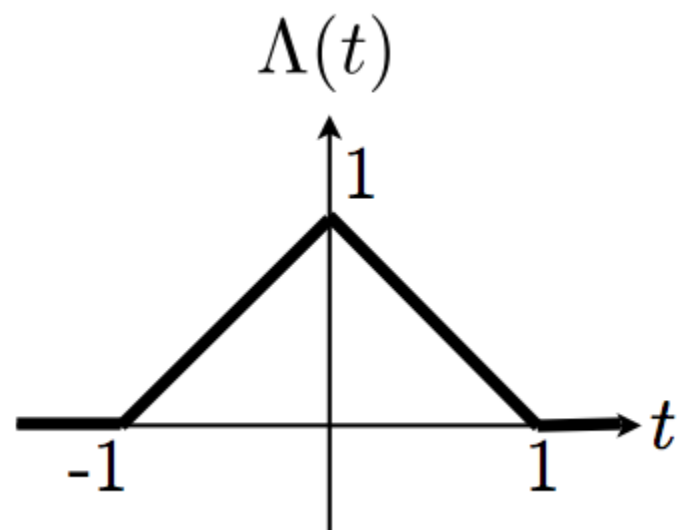
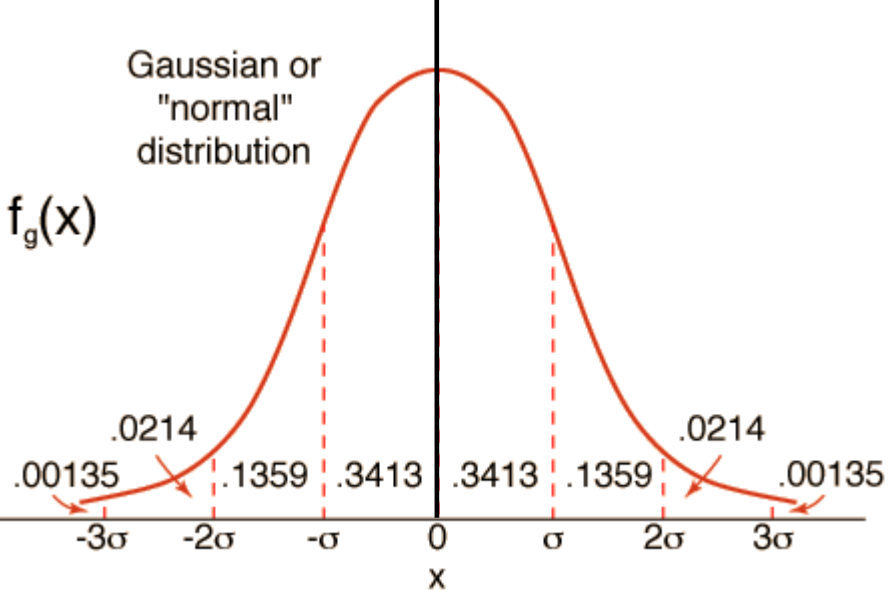
— an infinitesimally sharp pulse with an unity area.

Construction of the Unit-impulse function $\delta(t)$

One of the simplest way — rectangular pulse, taking the limit $\Delta \rightarrow 0$.

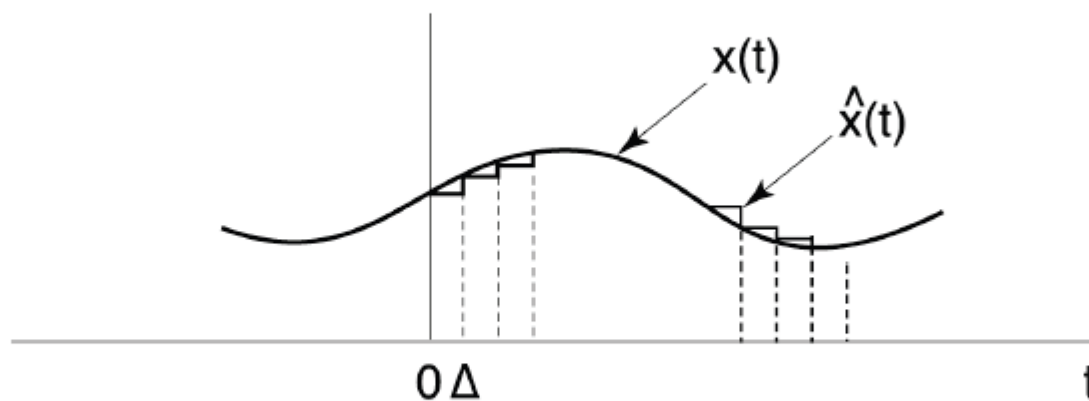


But this is by no means the only way. One can construct a $\delta(t)$ function out of many other functions, *Eg.* Gaussian pulses, triangular pulses, sinc functions, *etc.*, as long as the pulses are short enough — much shorter than the characteristic time scale of the system.



Representation of CT Signals

- Approximate any input $x(t)$ as a sum of shifted, scaled pulses (in fact, that is how we do integration)



$$\hat{x}(t) = x(k\Delta), \quad k\Delta < t < (k+1)\Delta$$

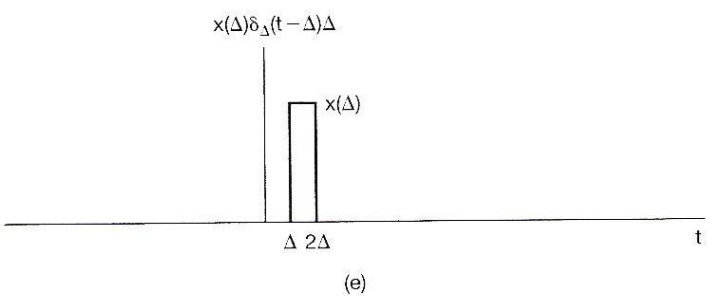
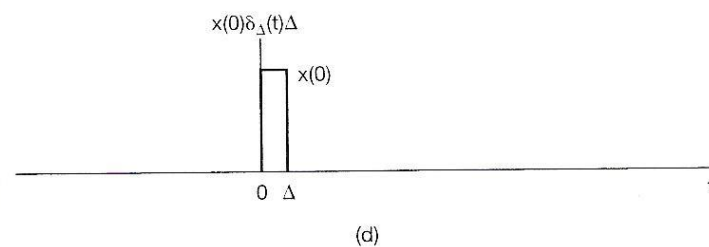
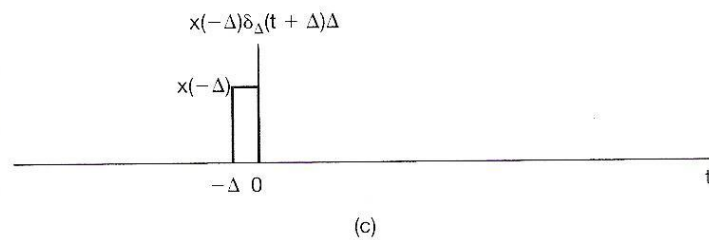
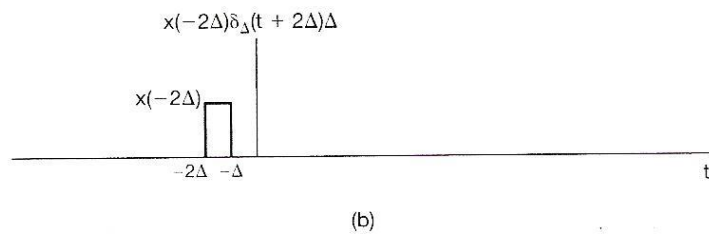
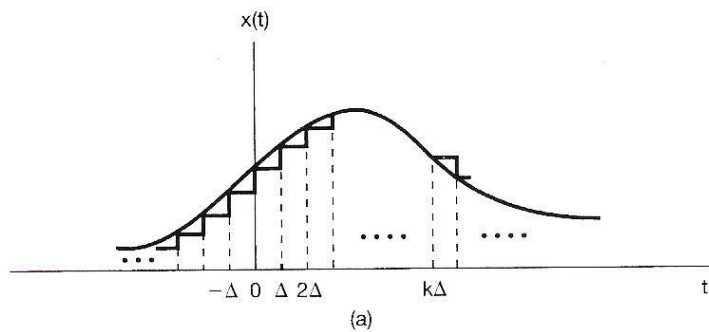
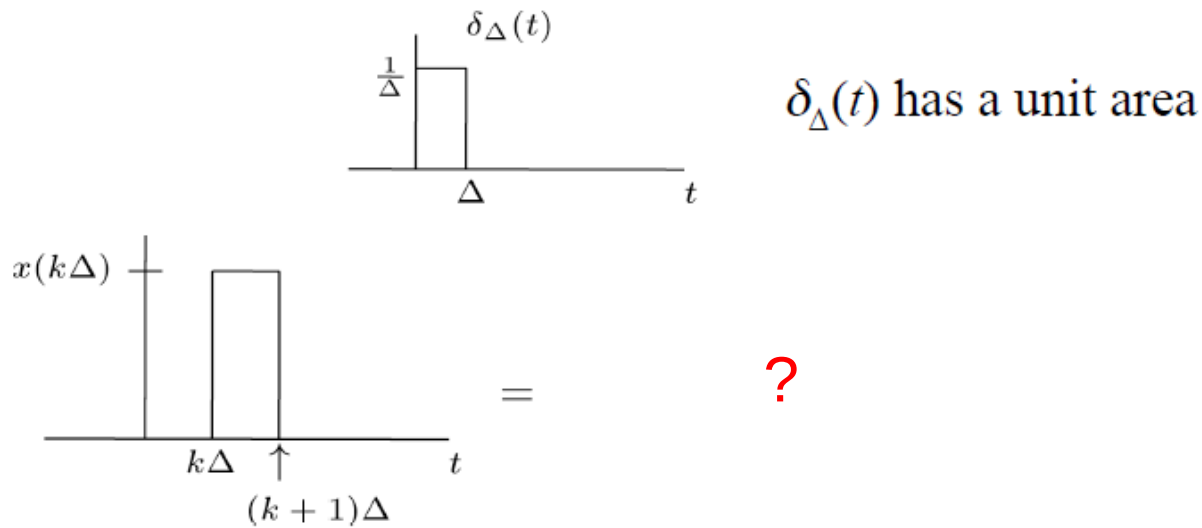


Figure 2.12 Staircase approximation to a continuous-time signal.

Representation of CT Signals (cont.)



$$\hat{x}(t) = \sum_{k=-\infty}^{+\infty} x(k\Delta) \delta_{\Delta}(t - k\Delta) \Delta$$

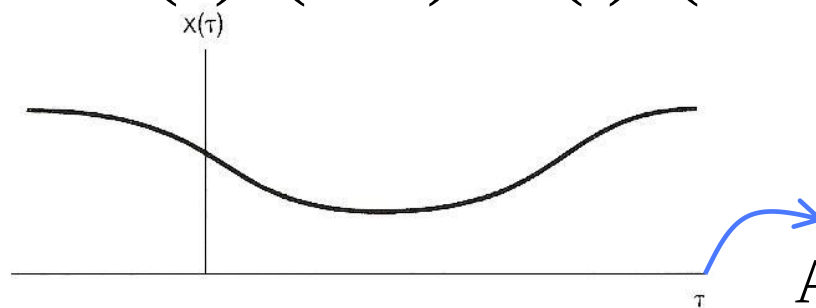


limit as $\Delta \rightarrow 0$

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

Sifting
property
of the unit
impulse

Recall, we have $x(\tau)\delta(t-\tau) = x(t)\delta(t-\tau)$, as a function of τ with t fixed



A function of τ with t fixed

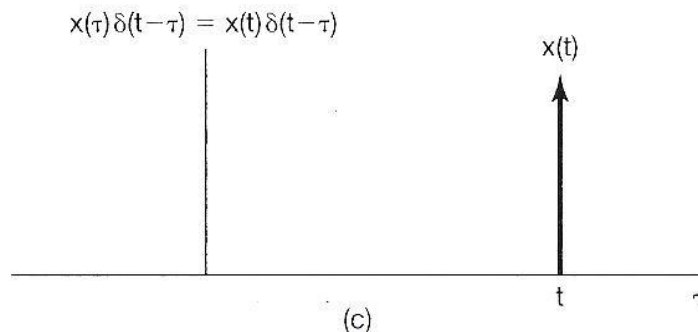
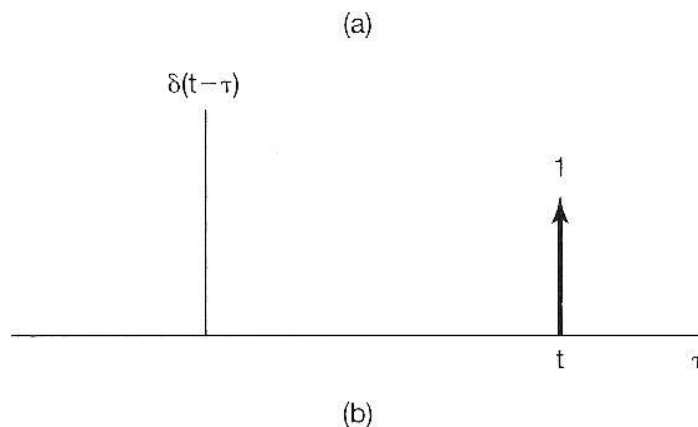
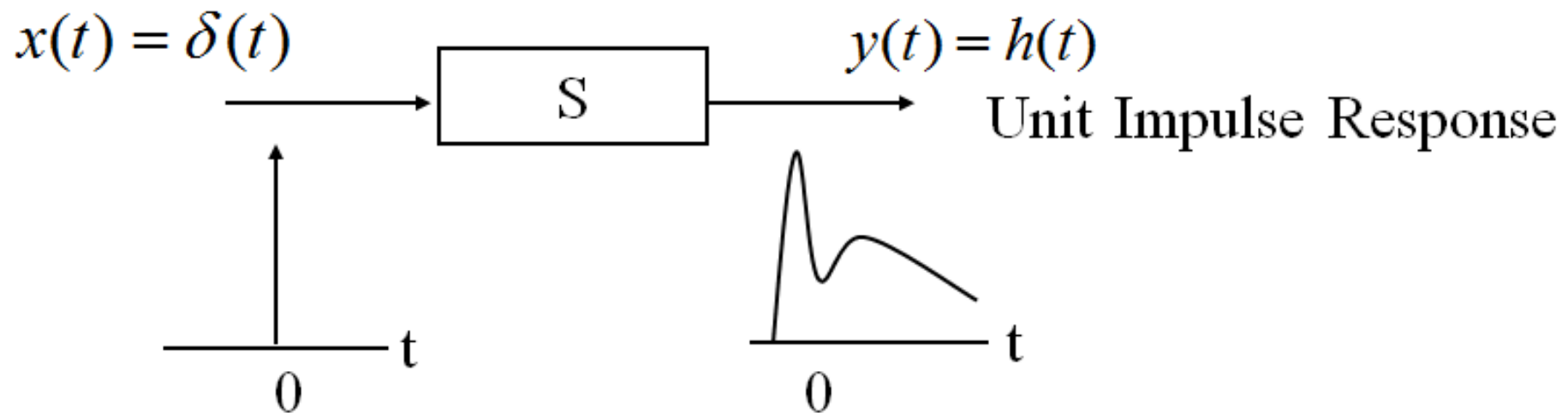


Figure 2.14 (a) Arbitrary signal $x(\tau)$; (b) impulse $\delta(t-\tau)$ as a function of τ with t fixed; (c) product of these two signals.

$$\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau = \int_{-\infty}^{\infty} x(t)\delta(t-\tau)d\tau = x(t)\int_{-\infty}^{\infty} \delta(t-\tau)d\tau = x(t)$$

Unit Impulse Response



Response of a CT LTI System



- Now suppose the system is **LTI**, and define the *unit impulse response* $h(t)$:

$$\delta(t) \longrightarrow h(t)$$



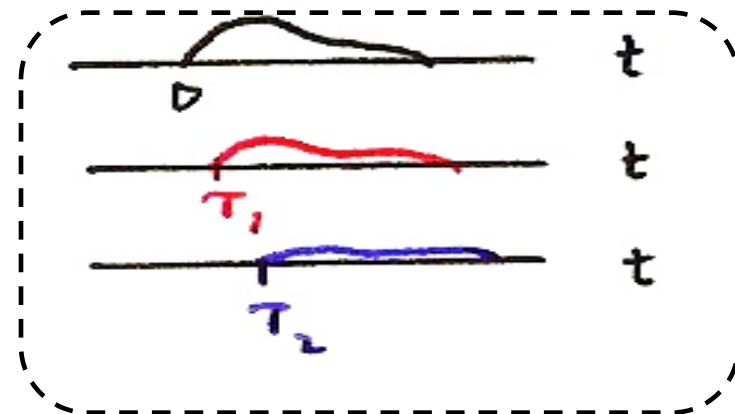
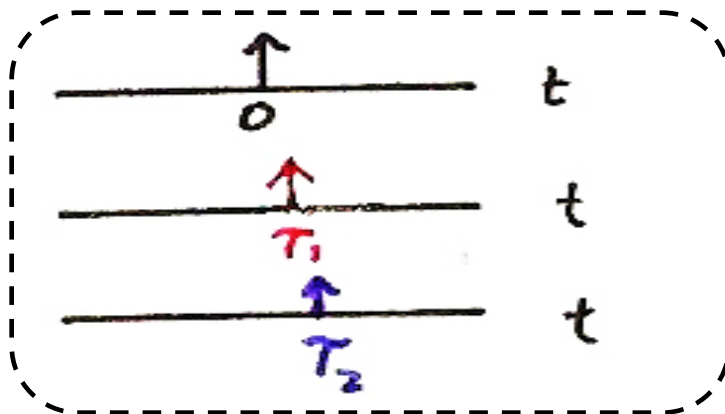
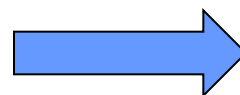
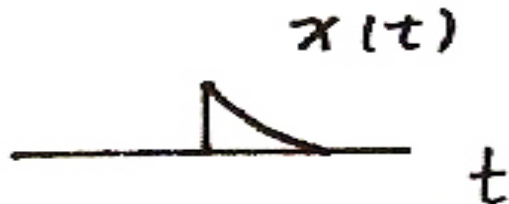
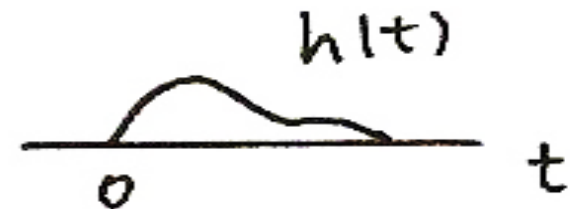
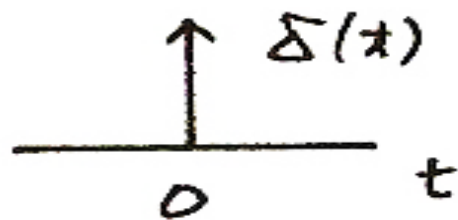
From **T**ime-**I**nvariance:

$$\delta(t - \tau) \longrightarrow h(t - \tau)$$

From **L**inearity:

$$x(t) = \int_{-\infty}^{+\infty} x(\tau) \delta(t - \tau) d\tau \longrightarrow y(t) = \underbrace{\int_{-\infty}^{+\infty} x(\tau) h(t - \tau) d\tau}_{\text{Convolution Integration}} = x(t) * h(t)$$

Response of a CT LTI System (cont.)





Important

Operation of CT Convolution

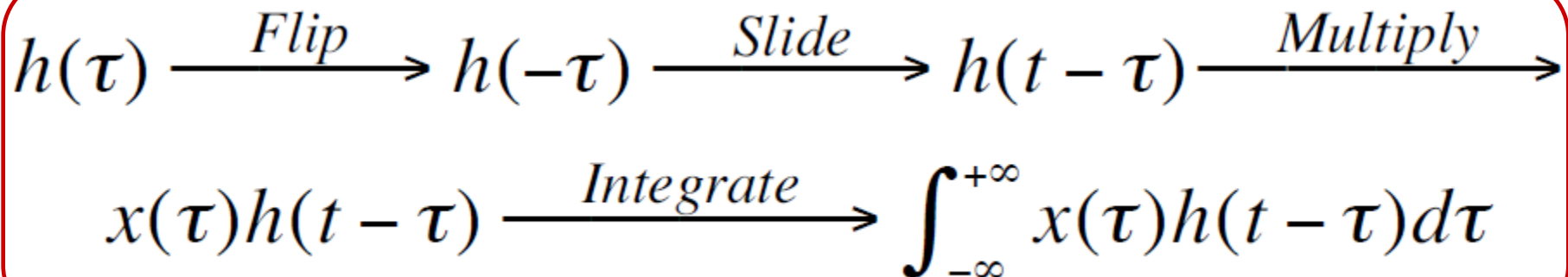
$$y(t) = x(t) * h(t) \equiv \int_{-\infty}^{+\infty} x(\tau) h(t - \tau) d\tau$$

Convolution Integral

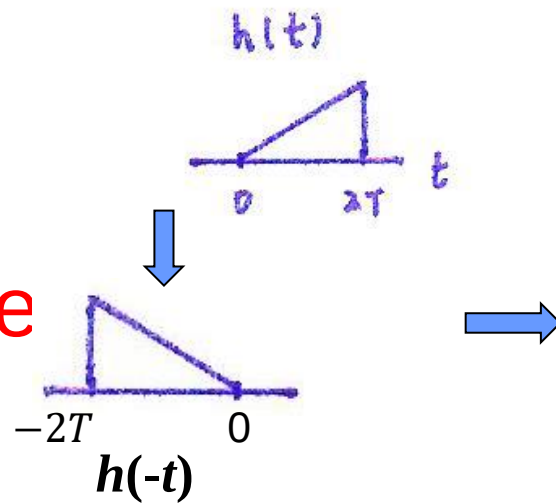
$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

Convolution Sum

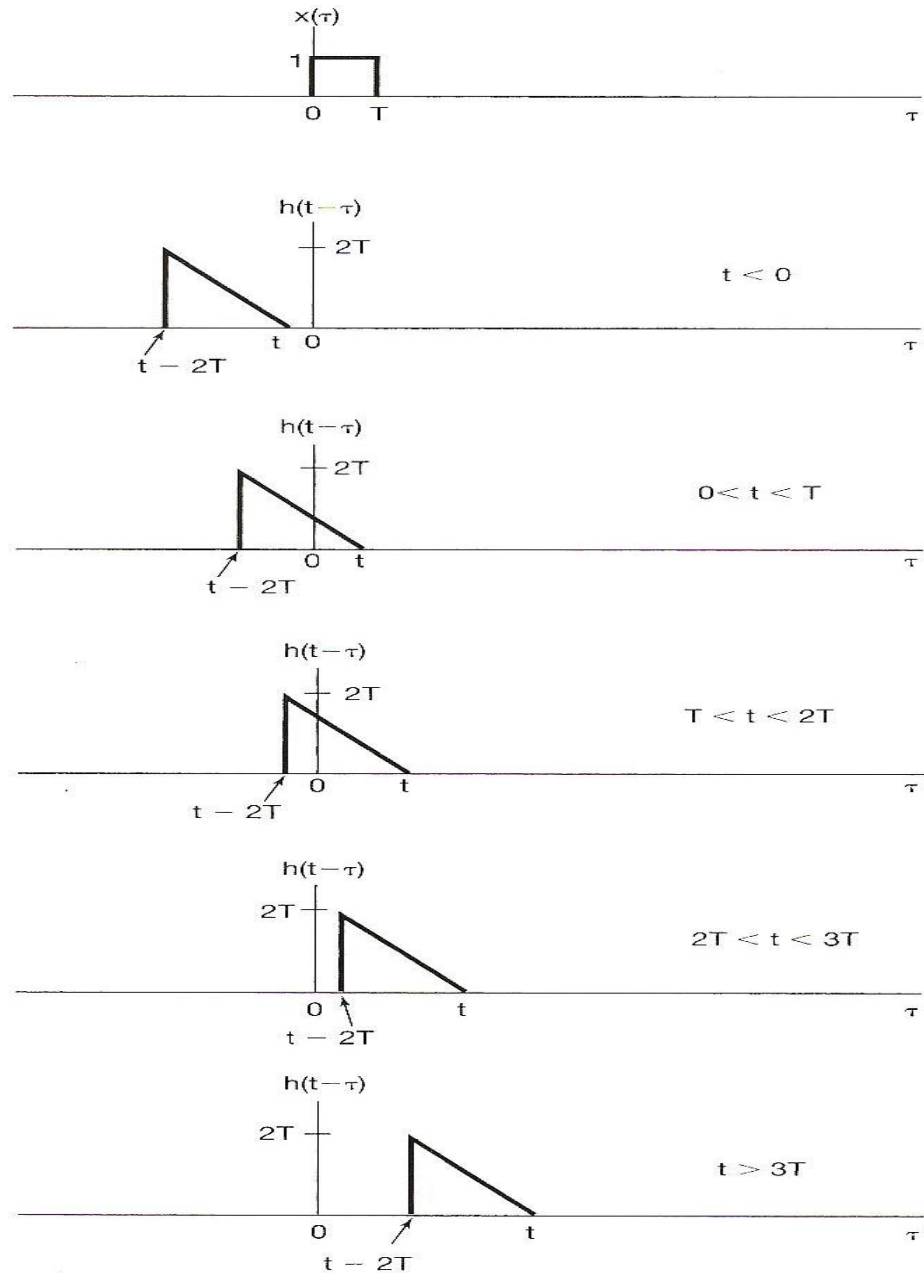
- A different way to understand the convolution integral $y(t)$ is a **weighted integral of the input**, where the **weight on $x(\tau)$ is $h(t - \tau)$**



flip, slide



⋮

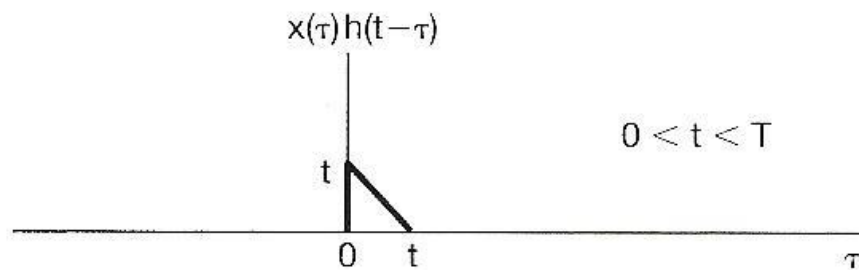


Sig

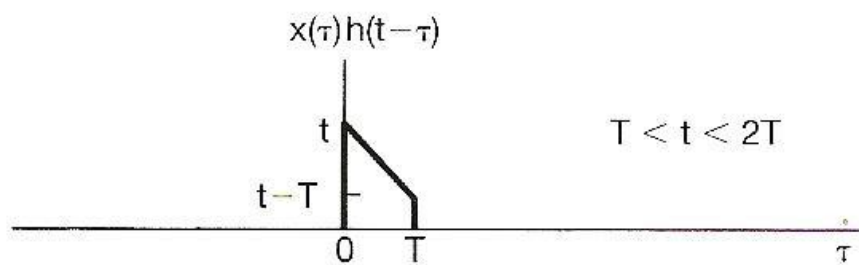
Figure 2.19 Signals $x(\tau)$ and $h(t-\tau)$ for different values of t for Example 2.7.

multiply

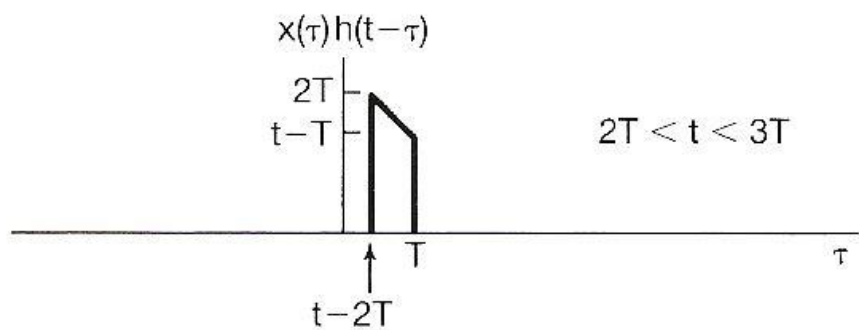
$$x(\tau)h(t - \tau)$$



(a)



(b)



(c)

Figure 2.20 Product $x(\tau)h(t - \tau)$ for Example 2.7 for the three ranges of values of t for which this product is not identically zero. (See Figure 2.19.)

integrate

$$\int_{-\infty}^{+\infty} x(\tau)h(t - \tau)d\tau$$

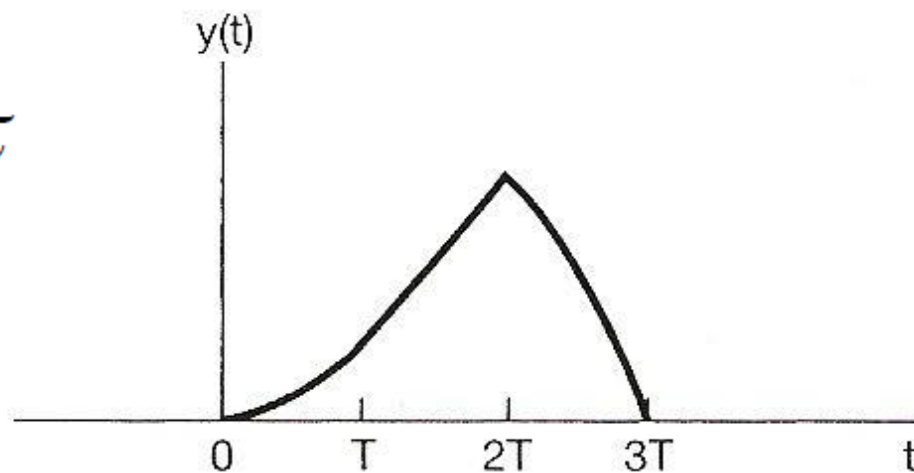
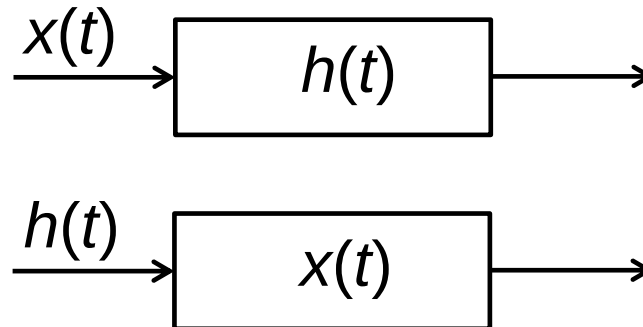


Figure 2.21 Signal $y(t) = x(t) * h(t)$ for Example 2.7.

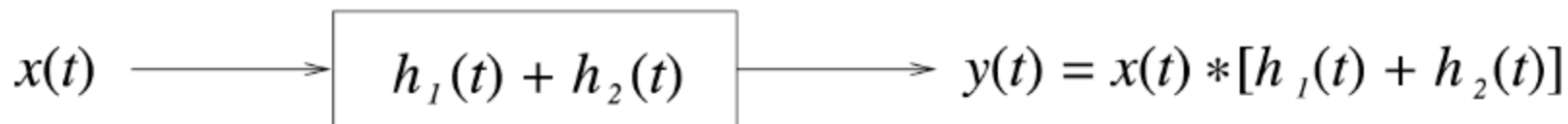
Property: Commutative

$$x(t) * h(t) = h(t) * x(t)$$

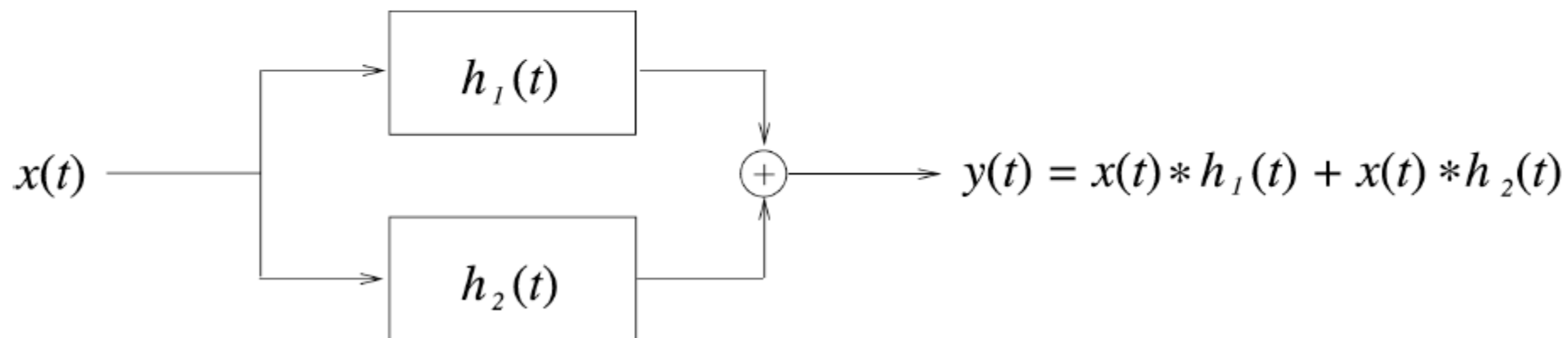
- The role of input signal and unit impulse response is **interchangeable**, giving the same output signal



Property: Distributive



||



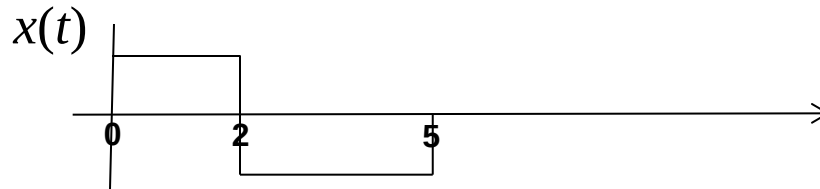
Property: Distributive (Cont.)

$$[x_1(t) + x_2(t)] * h(t) = x_1(t) * h(t) + x_2(t) * h(t)$$

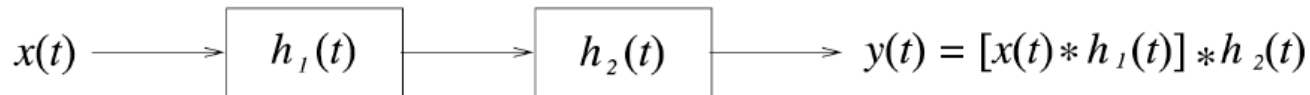
Problem 2.22 (b)

$$x(t) = u(t) - 2u(t - 2) + u(t - 5)$$

$$h(t) = e^{2t}u(1-t)$$



Properties: Associative



||

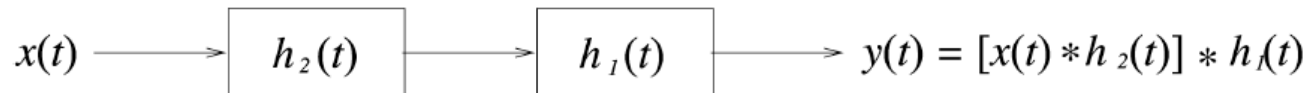


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← Commutativity



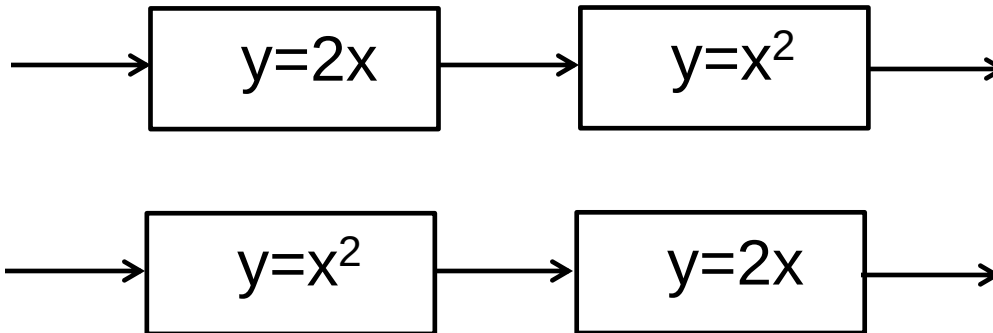
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- Cascade of two systems gives an unit impulse response which is the **convolution of the unit impulse responses** of the two individual systems
- The behavior of a cascade of two systems is **independent** of the order in which the two systems are cascaded

Properties: Associative (Cont.)

- The order in which non-linear systems are cascaded cannot be changed.
- e.g.



Property: Memory/Memoryless

- A linear, time-invariant, causal system is memoryless only

$$\text{if } h[n] = K\delta[n] \quad h(t) = K\delta(t)$$

$$y[n] = Kx[n] \quad y(t) = Kx(t)$$

if $K=1$ further, they are identity systems

$$y[n] = x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] = x[n] * \delta[n]$$

$$y(t) = x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau = x(t) * \delta(t)$$

Property: Invertibility

2.50. Consider the cascade of two systems shown in Figure P2.50. The first system, A , is known to be LTI. The second system, B , is known to be the inverse of system A . Let $y_1(t)$ denote the response of system A to $x_1(t)$, and let $y_2(t)$ denote the response of system A to $x_2(t)$.

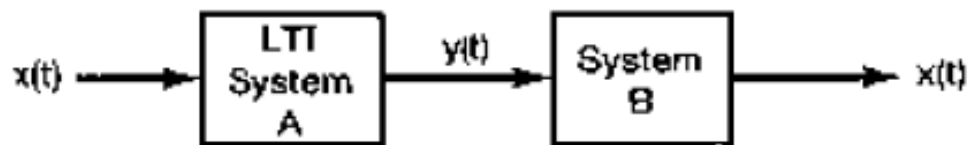


Figure P2.50

- (a) What is the response of system B to the input $ay_1(t) + by_2(t)$, where a and b are constants?
- (b) What is the response of system B to the input $y_1(t - \tau)$?

Property: Causality

Causality: CT LTI system is causal $\Leftrightarrow h(t) = 0$, at $t < 0$

- This is because that the input unit impulse function $\delta(t)=0$ at $t < 0$

As a result:

$$y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t - \tau)d\tau = \int_{-\infty}^t x(\tau)h(t - \tau)d\tau$$

$t - \tau \geq 0$, or $\tau \leq t$

$y(t)$ only depends on $x(\tau < t)$.

Property: Stability

BIBO Stability: CT LTI system is stable $\Leftrightarrow \int_{-\infty}^{+\infty} |h(\tau)| d\tau < \infty$

→ Sufficient condition:

For $|x(t)| \leq x_{\max} < \infty$,

$$|y(t)| = \left| \int_{-\infty}^{+\infty} x(\tau) h(t - \tau) d\tau \right| \leq x_{\max} \left| \int_{-\infty}^{+\infty} h(t - \tau) d\tau \right| < \infty.$$

→ Necessary condition:

Suppose $\int_{-\infty}^{+\infty} |h(\tau)| d\tau = \infty$

Let $x(t) = h^*(-t)/|h^*(-t)|$, then $|x(t)| \equiv 1$ bounded

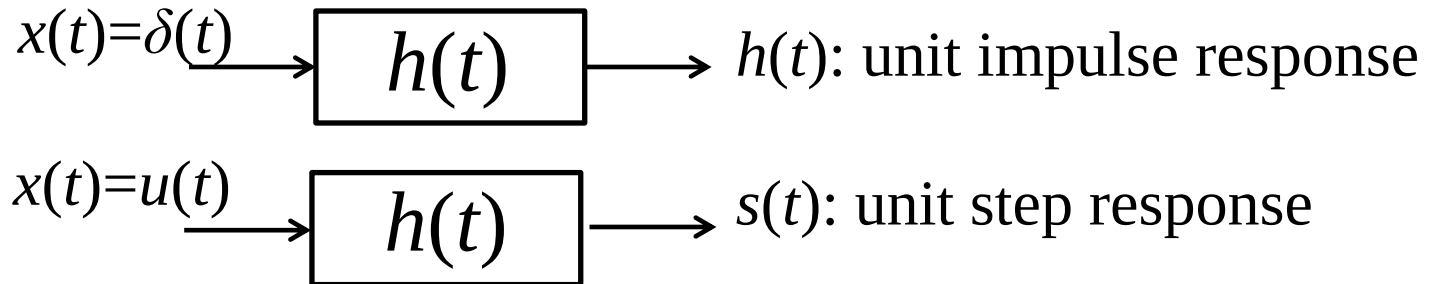
$$\text{But } y(0) = \int_{-\infty}^{+\infty} x(\tau) h(-\tau) d\tau = \int_{-\infty}^{+\infty} \frac{h^*(-\tau) h(-\tau)}{|h(-\tau)|} d\tau = \int_{-\infty}^{+\infty} |h(-\tau)| d\tau = \infty$$

Property: Unit Step Response

unit step function $\rightarrow$ unit step response

Step response

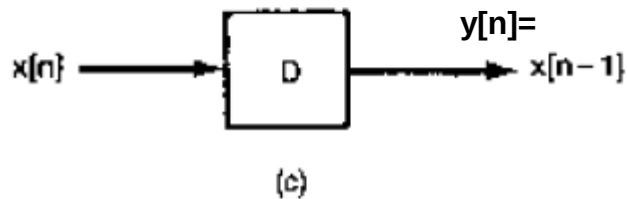
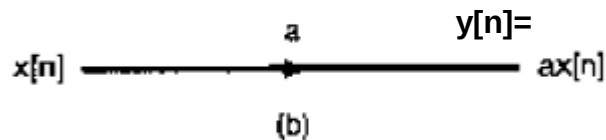
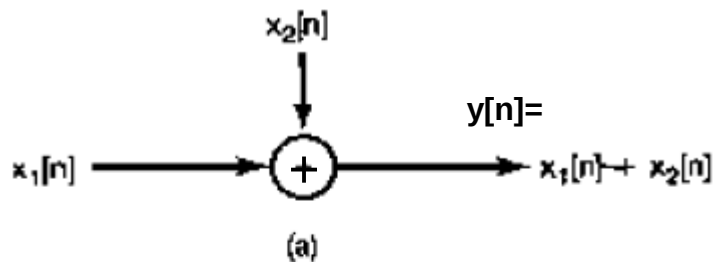
$$s(t) = u(t) * h(t) = h(t) * u(t) = \int_{-\infty}^t h(\tau) d\tau$$



The **relation** between unit step function and unit impulse function

$$h(t) = \frac{ds(t)}{dt} = s'(t)$$

Block diagram representation of 1-st order system



Problem 2.38

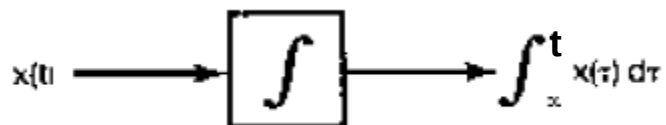
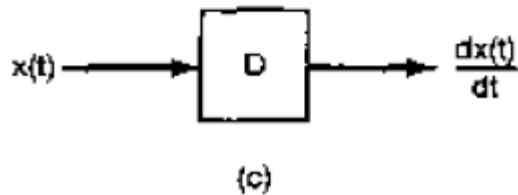
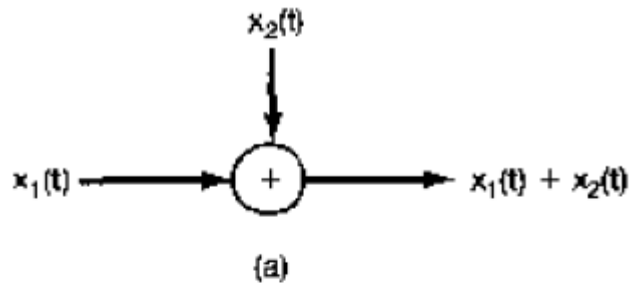
$$y[n] = \frac{1}{3} y[n-1] + \frac{1}{2} x[n]$$

$$y[n] = \frac{1}{3} y[n-1] + x[n-1]$$

How about

$$y[n] = \frac{1}{3} y[n-1] + 2y[n-2] + x[n-1] + 3x[n-2]$$

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^N b_k x[n-k]$$

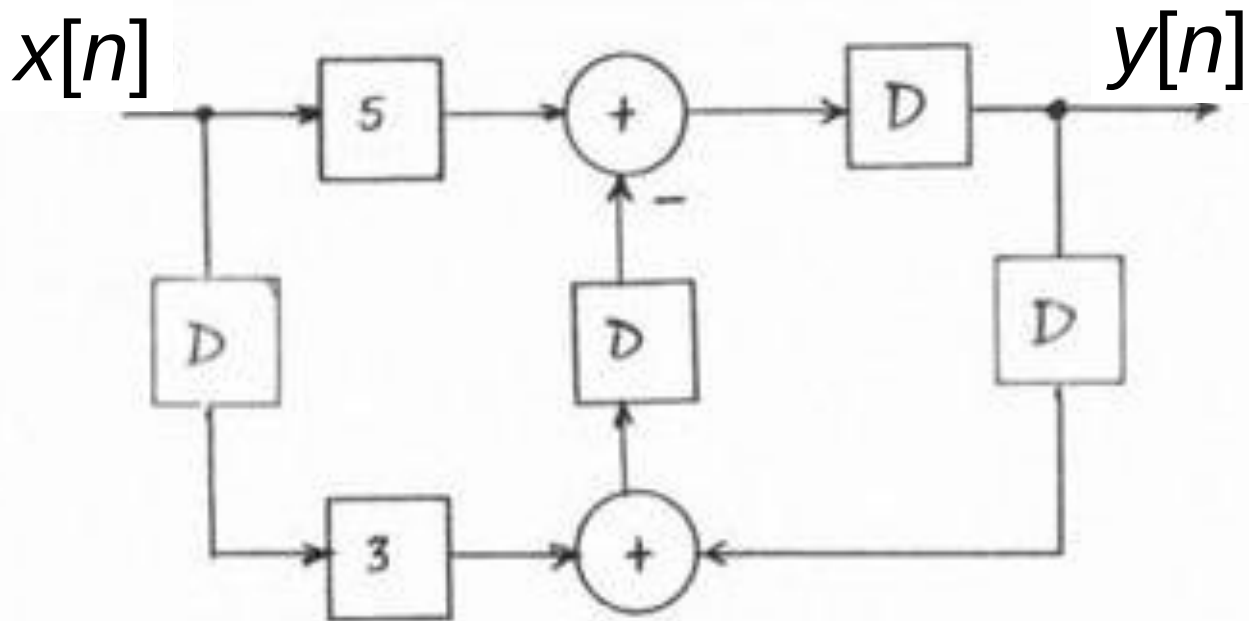


Problem 2.39

$$y(t) = -\frac{1}{2} \frac{dy(t)}{dt} + 4x(t)$$

$$\frac{dy(t)}{dt} + 3y(t) = x(t)$$

From block diagram to difference equation



More about $\delta(t)$: Operational Definition

A function can be defined by

- what it is at each **value** of the independent variable, or
- what it does under some mathematical **operation** (such as an integral or a convolution), or how it behaves with a system, or some mathematical constraints: **Singularity Function**

Operational Definition of Unit Impulse

- $\delta(t)$ can be defined as

- $x(t) = x(t) * \delta(t)$ for any $x(t)$, $x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$

- a signal which, when applied to a system, yields the impulse response $h(t) = h(t) * \delta(t)$

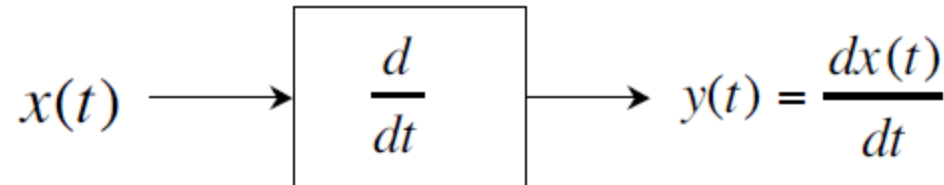
- such definition leads to, or is equivalent to, other properties of $\delta(t)$, $\int_{-\infty}^{\infty} \delta(\tau) d\tau = 1$

$$\int_{-\infty}^{\infty} g(\tau) \delta(\tau) d\tau = g(0)$$

they are also “operational definition” of $\delta(t)$

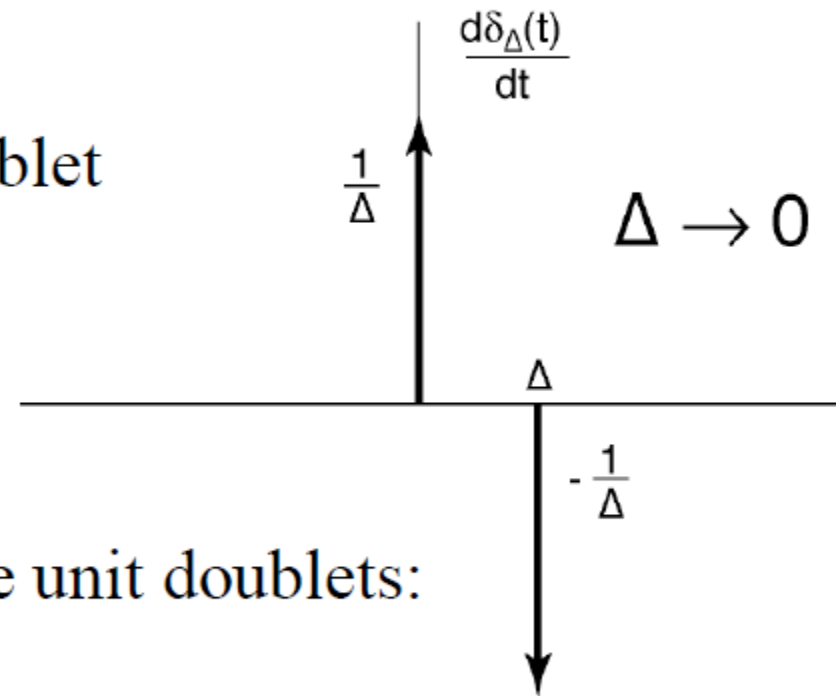
- Such definition also leads to the sampling property $f(t) \delta(t) = f(0) \delta(t)$

The Unit Doublet — Differentiator



Impulse response = unit doublet

$$u_1(t) = \frac{d\delta(t)}{dt}$$



The operational definitions of the unit doublets:

$$x(t) * u_1(t) = \frac{dx(t)}{dt}$$

Triplets and beyond!

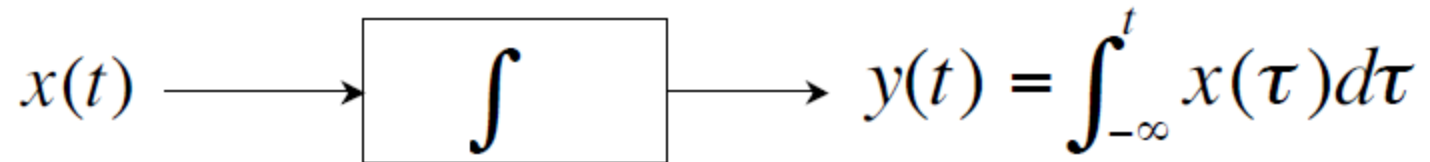
$$n > 0$$

$$u_n(t) = \underbrace{u_1(t) * \cdots * u_1(t)}_{n \text{ times}}$$

Operational definitions:

$$x(t) * u_n(t) = \frac{d^n x(t)}{dt^n} \quad (n > 0)$$

Integrators



Impulse response:

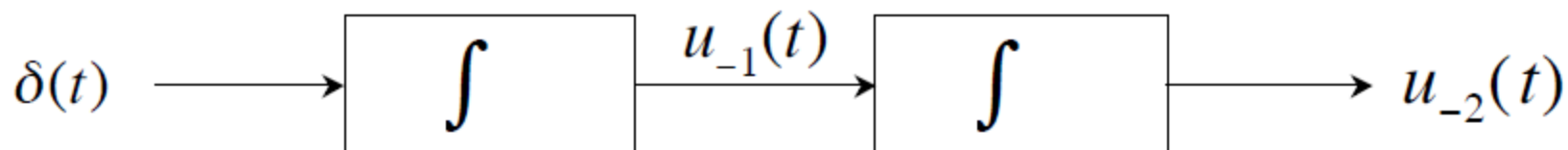
$$u_{-1}(t) \equiv u(t)$$

Operational definition: $x(t) * u_{-1}(t) = \int_{-\infty}^t x(\tau) d\tau$

Cascade of n integrators:

$$u_{-n}(t) = \underbrace{u_{-1}(t) * \cdots * u_{-1}(t)}_{n \text{ times}} \quad (n > 0)$$

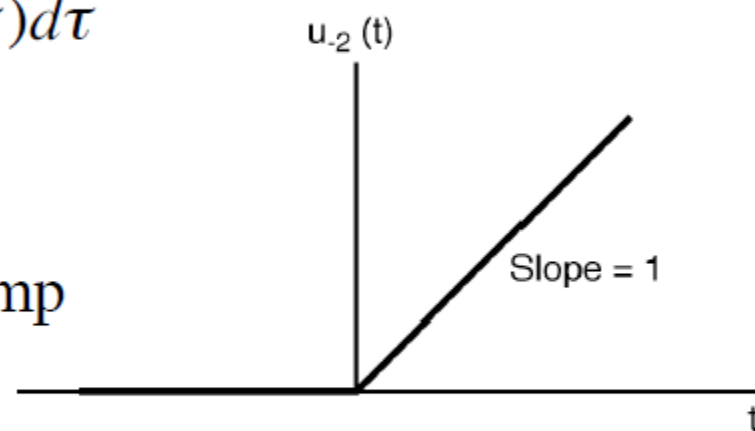
Integrators (Cont.)



$$u_{-2}(t) = \int_{-\infty}^t u_{-1}(\tau) d\tau = \int_{-\infty}^t u(\tau) d\tau$$

$$= u(t) \int_0^t d\tau$$

$$= t \cdot u(t) \quad \text{the unit ramp}$$



More generally, for $n > 0$

$$u_{-n}(t) = \frac{t^{(n-1)}}{(n-1)!} u(t)$$

Notation

Define

$$u_0(t) = \delta(t)$$

Then

$$u_n(t) * u_m(t) = u_{n+m}(t)$$

n and m can be $\pm$

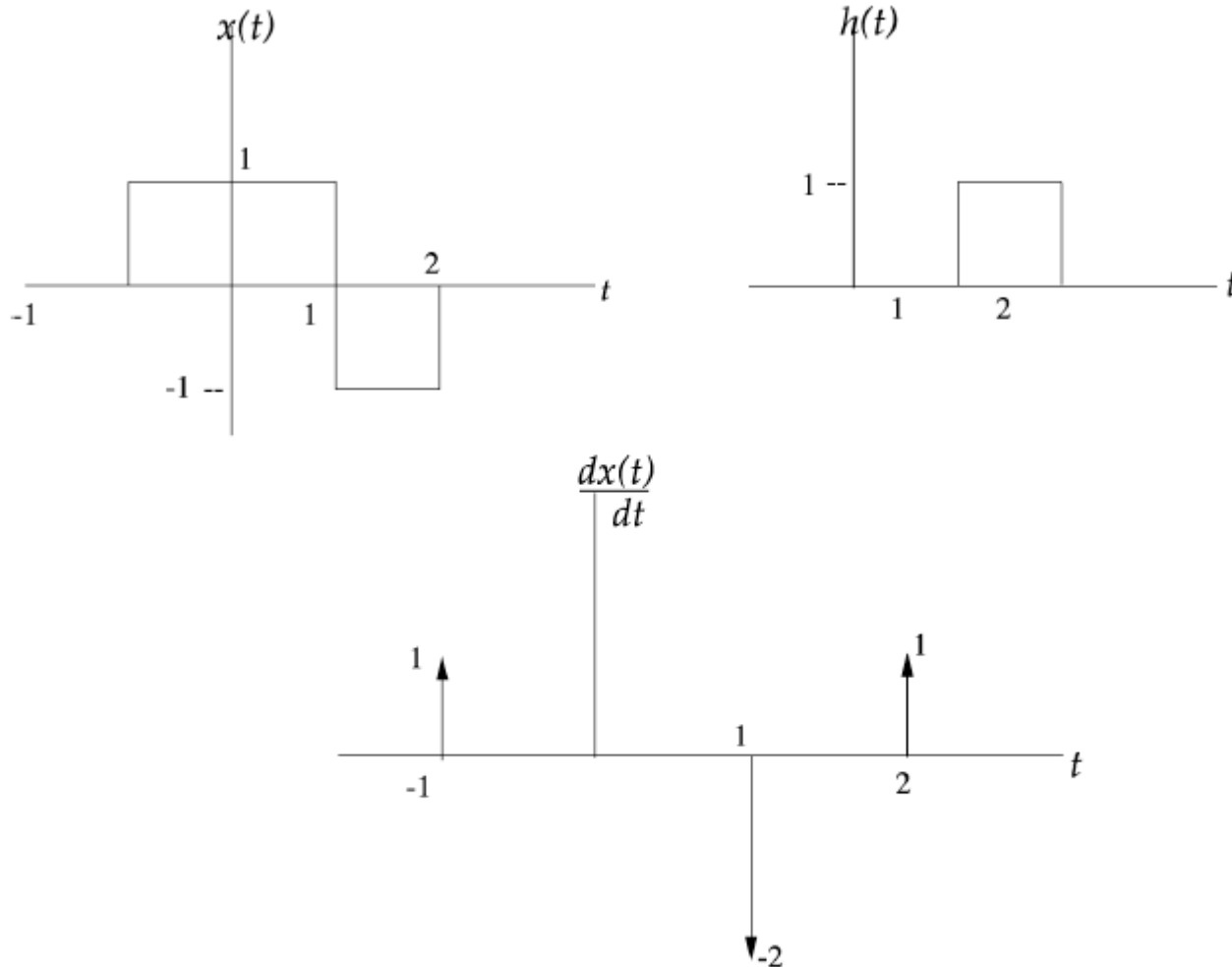
E.g.

$$u_1(t) * u_{-1}(t) = u_0(t)$$

||

$$\left(\frac{d}{dt} u(t) \right) = \delta(t)$$

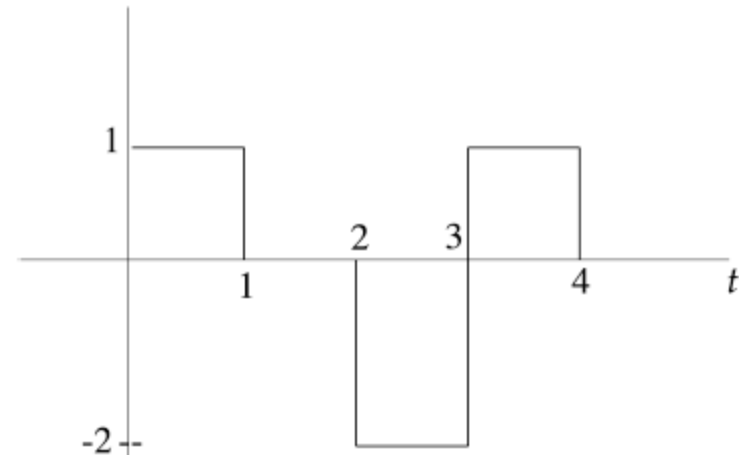
Example: Calculating $x(t)*h(t)$



$$\frac{dx(t)}{dt} = \delta(t + 1) - 2\delta(t - 1) + \delta(t - 2)$$

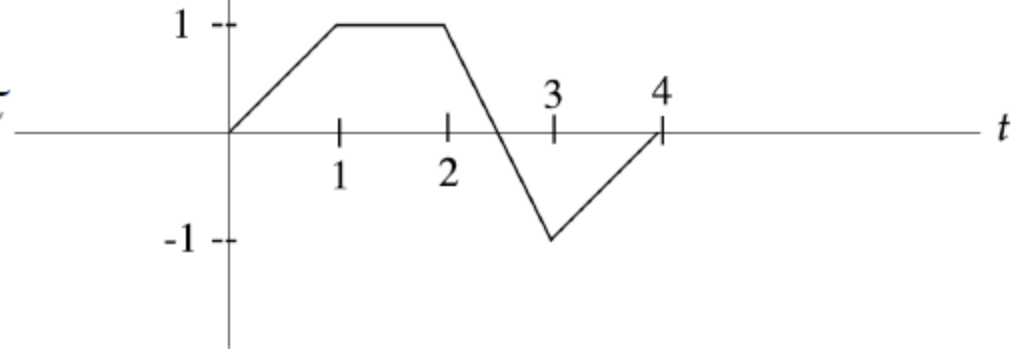
Example (Cont.)

$$\frac{dx(t)}{dt} * h(t) = h(t+1) - 2h(t-1) + h(t-2)$$



$$x(t) * h(t) = u_{-1}(t) * \{[u_1(t) * x(t)] * h(t)\}$$

$$= \int_{-\infty}^t \left[\frac{dx(\tau)}{d\tau} * h(\tau) \right] d\tau$$



Summary

- **How to represent a CT signal with approximation of DT signals**
- **Calculation of CT convolution**
- **Block diagram representation of a LTI system**
- **Operational definition of unit impulse function**